The Cohesion UFT Research Calculator, Version 6.5

The Recursion Unit Identified, and the Sixth Problem Reposed

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Cohesion UFT series · Research Calculator · October 2026

Abstract

Version 6.5 of the Cohesion UFT Research Calculator changes no number of the record. It carries two changes of standing that the series made after Version 6.4 [1]. First, the recursion unit in seconds, which Version 6.4 printed on Module 3 as two readings, $2/\omega_C = 2.6 \times 10^{-21}$ s and $2/(g\omega_C) = 6.2 \times 10^{-19}$ s, marked open between them, is identified from two closures the record already held [2, 3]: the electron closure of E = pr fixes the recursion length at λ_C, the trapping condition fixes the rest energy at $\hbar\omega_C$, and the medium's rate c joins them, so that the unit is the radian time of the minimal trapped recursion, $u_e = \lambda_C/c = \hbar/(m_e c^2) = 1.2881 \times 10^{-21}$ s. The two readings were two counts of this one unit. Module 3 now prints the unit once, at the standing identified, and the cadence of the fine-structure chain in seconds, $t_{coh} = 478.75\ u_e = 6.167 \times 10^{-19}$ s, with its rate and its restart acceleration. Second, the sixth problem of the series, posed as the torsion-interval energy fraction $E_{switch} = f{\cdot}m_e c^2$ on the c-bit paper's bound [4, 5], is reposed on Module 15 at the lattice scale and the bound withdrawn, because the measured spin flip of a single free electron, 0.6 meV at 5.4 T and falling with the field, is against it; the problem is now the torsion slope at the lattice scale, with the exchange energy per spin of a ferromagnet as its target and the Zeeman flip as the free-electron limit a derivation must return. Module 3 prints the release $2\hbar g$ at the electron's rate, $2g{\cdot}m_e c^2 = 4.27$ keV, as the free recursion's gear-shift and not the switching energy of a device. Nothing is fitted. The revised standalone calculator accompanies this paper.

1. What Version 6.4 Left Open

Version 6.4 completed the record's printed numbers on their modules and rebased the open programme on the problem paper [4]. Two of its cards carried the standing open where the series has since moved. On Module 3, the forward card printed the recursion clock in the derived torsion slope's own units, $t_{coh} = 2/g = 478.75$ units with $\nu_r = g/2$, $a_r = g^2/\omega$ [11] and the per-peak slip $\Delta\theta/6$ [12], and then the unit in seconds as the textbook's third version carried it: fixed only by dimensional analysis through the Compton frequency, and left between two readings, the coherence interval taken as two units of $1/\omega_C$, 2.6×10^{-21} s, and the cadence taken as 2/g units of the same, 6.2×10^{-19} s. On Module 15, the sixth card read "bound stated; fraction open": the switching energy of a torsion-phase bit as a fraction f of the electron's rest energy, with f to be derived from the intra-level phase separation of the n = 2 mode, held against measurement once one existed.

The series closed the first and reposed the second in the interval between the versions: the first in the note *The Recursion Unit* [3] and the textbook's fourth version, Section 7.4 [2]; the second in the textbook's fourth version, Section 22.4, with its Appendices C and D.

Version 6.5 carries both onto the modules, with the arithmetic that follows from them and nothing else.

2. The Recursion Unit on Module 3

The interval $t_{coh} = 2/g$ is stated in recursion units with $\omega = 1$: the unit is the time in which the recursion advances one radian at its rate. Fixing it in seconds needs the rate of a recursion the record has closed, and the record has closed two things at the electron. The closure $E = pr$ holds at the electron with unit coefficient when the radius is the reduced Compton length [6], which fixes the recursion length, $\lambda_C = \hbar/(m_e c)$; and the trapping condition that fixes the electron as the $n = 2$ state [2, Ch. 11] gives its rest energy as $\hbar\omega_C$, the recursion slipping at the Compton angular rate $\omega_C = m_e c^2/\hbar = 7.7634 \times 10^{20}\ s^{-1}$. The medium's rate relates the two, since a rotation of radius λ_C at ω_C has boundary speed $\lambda_C\omega_C = c$: the minimal trapped recursion is the one whose boundary circulates at the medium's own rate. Its radian time is the unit.

$$u_e = \lambda_C/c = \hbar/(m_e c^2) = 1/\omega_C = 1.2881 \times 10^{-21}\ s$$

The two readings of Version 6.4 were two counts of this unit. The first, $2/\omega_C$, counts two units: it is the coherence limit $\tau(t_{coh}) = 2$ rad read as a time, the superseded $2/\omega$ form of the cadence. The second, $2/(g\omega_C)$, counts $2/g$ units: it is the interval of the recursion whose slope g is, the form the calculator has used since the forward chain was built. The record had already chosen between them, so the forward card now prints the unit once and the cadence in seconds after it. No ratio in the chain moves; $1/\alpha = 137.029$ is returned as before.

Row on the forward card	Printed value	Expression	Standing
$\omega_C = m_e c^2/\hbar$	$7.7634 \times 10^{20}\ s^{-1}$	the trapping condition's rate; the closure's length over the medium's rate, c/λ_C	CODATA inputs
Recursion unit $u_e = \lambda_C/c$	1.2881×10^{-21} s	$\hbar/(m_e c^2) = 1/\omega_C$	identified from two closures [3; 2, §7.4]
t_{coh} in seconds	6.1667×10^{-19} s	$(2/g)\cdot u_e$; 76.2 turns of the recursion per slip	derived count in the identified unit
ν_r	$1.6216 \times 10^{18}\ s^{-1}$	$g\omega_C/2$	derived count in the identified unit
a_r	1.0518×10^{37} rad/s^2	$g^2{\omega_C}^2$	derived count in the identified unit
$2\hbar g$ at the electron's rate	4.27 keV	$2g\cdot m_e c^2 = \Delta\theta\cdot m_e c^2$	form derived [2, §8.4]; no measurement attached

Two things about the card's standings. The unit is the one quantity on the forward card at the standing identified rather than derived, because it rests on the electron mass as a measured input; it becomes derived when the electron mass is derived from the axiom, which the textbook's Appendix C carries as the absolute scale of the cascade. And the last row is printed for what it is. The toggle releases $E_{slip} = \hbar\omega\cdot(2g/\omega) = 2\hbar g$ at the gear-shift [2, §8.4]; at the electron's rate this is $2g\cdot m_e c^2 = 4.27$ keV, the free recursion's own gear-shift, a

number once the unit is, with no measurement attached. It is not the switching energy of any device, for the reason Section 3 gives.

The identification instances a rule: the recursion unit of a scale is its coherence length over its propagation rate. At the graphene lattice the rule returns the clock the series identified there, $a/v_F = 2.46 \times 10^{-16}$ s [7], so a torsion slope derived at any scale becomes a time at once. The calculator does not compute the rule at scales other than the electron, because no slope is derived at any other scale; the C.2 card on Module 15 now says so.

3. The Sixth Problem Reposed on Module 15

The c-bit paper [5] stated a bound: that the switching energy of a torsion-phase bit must exceed the thermal floor $k_B T \approx 26$ meV at room temperature, that the torsion-interval energy setting the phase separation of the two-pole mode is a fraction of the electron's rest energy, some four orders above that floor, and that the exact fraction awaits derivation. The problem paper posed the fraction as its sixth problem [4], and Versions 6.3 and 6.4 carried the card at "bound stated; fraction open".

The measurement the bound has to be held against exists, and it is against it. The c-bit paper reads the torsion phase as the orientation of the local moment, so that its NOT is a spin flip, and the spin flip of a single free electron is measured: in a Penning trap at 5.4 T it is driven at the Larmor frequency, about 150 GHz, 0.6 meV per flip [8], and the cost falls with the field toward nothing; paramagnetism is the same fact at room temperature, spins not locked to a lattice randomised thermally. There is no free-electron floor on flipping a torsion phase. What makes a magnetic bit thermally stable is the lattice's lock, the exchange coupling between neighbouring recursions, of order 0.01 to 1 eV per spin in a ferromagnet and 90 meV at the Curie point of iron, a quantity of the lattice scale. The calculator's own form says the same: the switch releases $2\hbar g$ at the scale whose slope is g, and a bit held in a lattice switches at the lattice's slope and the lattice's rate, the rate being the unit of Section 2 instanced there. The record derives g only at the electron.

The card is therefore rewritten as one problem, in the problem paper's form. **Target:** the switching energy as $2\hbar g$ at the slope and rate of the lattice that holds the bit, against the exchange energy per spin of a ferromagnet, with the Zeeman flip $\mu_B B$ as the free-electron limit a derivation must return; and g at the graphene scale against the identified 0.49 fs and 647 THz, which stand. **Derive:** the torsion slope at the lattice scale, with the lattice's coupling $\bar{I}$ in place of $3\alpha/2$; the unit is already the rule's, a/v_F. **Excluded:** a slope read off a measured switching energy; a lattice g fitted to the identified interval; a free-electron floor in any form. **Standing:** form derived; lattice slope open. The badge reads so, and the card records that the bound is withdrawn and why, so that a reader of the c-bit paper's first version finds the correction where the problem is evaluated.

The reposing changes the problem's place in the open list without changing its neighbours. It makes the sixth problem an instance of the fifth's route: a torsion slope at a scale above the electron, from that scale's geometry and coupling, which is the same derivation the fluid vortex, the cyclone, the pulsar and the graphene lattice wait on, and group C.2 names it as such. The frequency ceiling of the c-bit paper stands as stated: the torsion cycle of the electron's two-pole recursion is tied to the Compton frequency, $m_e c^2/h = 1.24 \times 10^{20}$ Hz, a geometric ceiling and not a device rate.

4. What Changed in the Calculator

Module	Version 6.4	Version 6.5
0	The printed numbers completed and Module 15 rebased	The recursion unit closed and Problem 6 reposed, with the sources at their current versions.
3	Forward card: the recursion clock in units, then "the unit in seconds" as two readings, open	Forward card: ω_C read as the two closures' rate; the unit $u_e = \lambda_C/c$, identified; t_{coh}, ν_r and a_r in seconds; $2\hbar g$ at the electron's rate, 4.27 keV, as the gear-shift's release and not a switching energy. The note cites the identification.
15	Problem 6: "bound stated; fraction open", $f \cdot m_e c^2$ as the target. C.2: the unit open between two readings	Problem 6: the switching energy at the lattice scale, "form derived; lattice slope open", the free-electron bound withdrawn with the measurement that withdraws it. C.2: the unit identified; the rule that turns a slope at any scale into a time.
16	The cadence row with the unit open between 2.6×10^{-21} s and 6.2×10^{-19} s	The cadence row in seconds; rows for the unit and for $2\hbar g$ at the electron's rate.
19	Sources note: the first version of the perihelion paper; the correction stated as R_s 0.14% low	Sources note cites the second version [9], which carries the corrected figure; the correction restated as the paper's table printed it: R_s = 2949.9 m, 0.11% low, which with the first version's rounded elements and orbit count returned 42.92″.
Header, footer	Version 6.4	Version 6.5; the description names the two changes.

Module 19's arithmetic is unchanged; the one-line restatement in its note separates the first version's radius error, 0.11% in R_s and 0.05″ in the result, from the rounding of its elements and orbit count, 0.01″, as the second version of the perihelion paper does. No other module changes in text or in code. The standard applied is the calculator's [10]: a displayed value must follow from its cited formula, nothing is adjusted toward observation, and the badge says what the computation establishes in the textbook's terms.

5. Re-execution

Every module was re-executed after the revision in a browser, and every value of Version 6.4 is reproduced, the executed record of Universal Mechanics [13] and the textbook's Appendix A among them: $1/\alpha = 137.029$ with the +0.007 identified, $\lambda_1 = 205.554$, $\lambda_2 = 16.357$, $\lambda_3 = 1.30$, $\Phi_t = 1.249570$, R = 0.1638 at $D/D_{GUT} = 10^{-4}$, the epoch ratio 1/33.2 against 1.083 with 1.614 required, $a_0 = 1.379 \times 10^{-10}$ m/s^2 at $H_0 = 70$, $D_{trans}/D_{GUT} = 0.439$, $\beta^{-1} = 4.26$ (km/s)^2/pc, the closure count 16 with both 3/8 checks, the mode-separation factor 0.708, the slope ratio 19.97, the proton closure 938.272 MeV, $\tau_p = 6.25 \times 10^{34}$ years, and 42.981″ per century for Mercury with 8.62″ for Venus and 3.84″ for Earth at their elements. The new rows of Module 3 return 1.2881×10^{-21} s, 6.1667×10^{-19} s, 1.6216×10^{18} s^{-1}, 1.0518×10^{37} rad/s^2, 76.2 turns and 4.27 keV from the constants the page already carried; no constant was added.

6. Propagation

The textbook [2]. The fourth version carries both changes: Section 7.4 identifies the unit, Section 22.4 withdraws the bound and reposes the problem, and Chapter 20's constraint table, the parameter sheet of Chapter 21 and Appendices C and D record them. Its module column for the recursion clock now points at rows that print the cadence in seconds; nothing in it needs to change for this version.

Problems for Derivation [4] and the Open problems page. The sixth problem's statement there is the first posing; its next version carries the problem as Module 15 now states it, and the page's standing for Problem 6 moves with it. The other six problems are unchanged.

The c-bit paper [5]. Its bound on the switching energy is withdrawn here and in the textbook; its second version withdraws it at the source, keeping the definition of the c-bit, the two states and the frequency ceiling, which stand.

The Research Calculator, Version 6.4 [1] and its companion paper. Superseded by this version; the 6.4 record remains available, and its completion of the printed numbers and rebasing of Module 15 stand.

cohesionuft.com. The site's calculator links resolve to the latest version through the record's concept DOI. The compact edition on the Calculator page carries neither the unit nor the sixth problem and is unchanged.

7. Quick Reference (Version 6.5 Additions)

Quantity	Value	Note
Recursion unit $u_e = \lambda_C/c$	1.2881×10^{-21} s	$\hbar/(m_e c^2) = 1/\omega_C$; identified from the electron closure and the trapping condition
$t_{coh} = 2/g$; $\nu_r = g/2$	$478.75\ u_e = 6.167 \times 10^{-19}$ s; $1.622 \times 10^{18}\ s^{-1}$	the slip cadence at the electron scale, in seconds through the identified unit
$a_r = g^2 \omega_C{}^2$	1.052×10^{37} rad/s^2	the mean restart acceleration in seconds
$2\hbar g$ at the electron's rate	4.27 keV	$2g \cdot m_e c^2$; the free recursion's gear-shift; not a switching energy
Problem 6	form derived; lattice slope open	the switching energy as $2\hbar g$ at the lattice's slope and rate; the free-electron bound withdrawn

8. Availability

The standalone calculator (Cohesion_UFT_Calculator_v6_5.html) is deposited with this paper. It runs in any browser with no installation; the charts load one library from a public CDN and every computation is self-contained.

References

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